

Asset Pricing Theory: S03

Dynamic Consumption and Portfolio Choice

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Structure of the course

- Email: amedeo.andriollo@dauphine.psl.eu
- Grading follows a 30-70 rule: 70% final exam, 30% project/homework (4th, 7th sessions).
- Office hours: feel “free” to (DO) come: B612.
- Slides/materials are updated regularly. Report a typo/mistake: corrections help the class.
- This session:
 - Continuous-time returns, Itô’s lemma, dynamic programming, the HJB, and the Merton consumption–portfolio problem.
 - Main references: Cochrane (2005, Sec. 1.5 and App. A); Campbell (2018, Chs. 2 and 9); Merton (1971).

⤵ Next session: Production-Based Asset Pricing.

Today's Route: Incorporating Dynamics

- Session 2 linked asset prices to marginal utility: $M_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)}$, and $1 = \mathbb{E}_t[M_{t+1} R_{i,t+1}^g]$.
- This session makes consumption and portfolio choice **jointly endogenous over time**.

Discrete time: finite-period returns

$$R_{t,t+1}^g = \frac{S_{t+1} + D_{t+1}}{S_t}.$$

- Payoffs and rebalancing occur over nonzero intervals.
- Gross returns compound over the interval.

Continuous time: local evolution

$$\frac{dS_t}{S_t} = \mu dt + \sigma dB_t.$$

- Decisions and shocks arrive continuously.
- Drift is order dt ; risk is order $\sqrt{dt}$.
- dS_t/S_t is an instantaneous simple return, not a discrete gross return.



The Investment Opportunity Set

- The investor can allocate wealth between a locally risk-free account A_t and a risky asset S_t .
- The money-market account earns the safe rate r (with its return is known at date t):

$$\frac{dA_t}{A_t} = r dt.$$

- The risky asset follows

$$\frac{dS_t}{S_t} = \mu dt + \sigma dB_t,$$

where μ is the expected instantaneous return, $\sigma > 0$ is volatility, and dB_t is the Brownian innovation.

- Over a short interval Δt , this implies approximately

$$\frac{\Delta S_t}{S_t} \approx \mu \Delta t + \sigma \sqrt{\Delta t} \varepsilon_{t+\Delta t}, \quad \varepsilon_{t+\Delta t} \sim N(0, 1).$$

Thus expected return accumulates at rate Δt , while uncertainty accumulates at rate $\sqrt{\Delta t}$.

- The compensation for bearing (local) risk is summarized by $\frac{\mu - r}{\sigma}$, i.e., the instantaneous Sharpe ratio (expected excess return per unit of volatility).

Diffusion Model

- A diffusion model has the form: $dX_t = a(X_t, u_t, t) dt + b(X_t, u_t, t) dz_t$, where X_t is the state variable and u_t is the control chosen using information available at date t .
- The drift term determines the mean, while the diffusion term b determines the scale (to the Brownian shock):

$$\mathbb{E}_t[dX_t] = a dt, \quad \text{Var}_t(dX_t) = b^2 dt.$$

- Brownian increments over a finite interval: $z_{t+\Delta} - z_t \sim N(0, \Delta)$.
↪ we write

$$z_{t+\Delta} - z_t = \sqrt{\Delta} \cdot \varepsilon_{t+\Delta}, \quad \varepsilon_{t+\Delta} \sim N(0, 1).$$

- Over a short interval,

$$\Delta X_t \approx a \Delta + b \sqrt{\Delta} \varepsilon_{t+\Delta}.$$

hence the predictable component is order Δ , while the random component is order $\sqrt{\Delta}$.

- As $\Delta \downarrow 0$, $\sqrt{\Delta} \gg \Delta$. Hence, over a very short interval, the (realized) changes in X_t are typically driven more by the random mean-zero shock than by the drift.
↪ Locally, uncertainty dominates what we **observe**, while the drift determines what we **expect**.

Why Does a Second-Order Term Survive?

- For a smooth function $f(X_t)$, a second-order Taylor expansion gives

$$df \approx f_X dX_t + \frac{1}{2} f_{XX} (dX_t)^2.$$

- Usually the quadratic term would be small. However, the random increment is of order $\sqrt{dt}$.
- If $dX_t = a dt + b dz_t$, then

$$(dX_t)^2 = a^2 (dt)^2 + 2ab dt dz_t + b^2 (dz_t)^2 = b^2 dt,$$

because

$$(dt)^2 = 0, \quad dt dz_t = 0, \quad \boxed{(dz_t)^2 = dt.}$$

- The key point is

$$dz_t = O(\sqrt{dt}) \implies (dz_t)^2 = O(dt).$$

So the quadratic variation of the Brownian shock survives at the same order as the drift.

→ Curvature therefore matters locally: for a nonlinear f , uncertainty contributes to the drift through

$$\frac{1}{2} b^2 f_{XX} dt.$$

Why Does a Second-Order Term Survive?

- For a smooth function $f(X_t)$, a Taylor expansion over a short interval: $\Delta f \approx f_X \Delta X + \frac{1}{2} f_{XX} (\Delta X)^2$.
- Suppose X follows the diffusion model: $\Delta X = a\Delta t + b\Delta z$, $\Delta z \sim N(0, \Delta t)$.
Since $\Delta z = \sqrt{\Delta t} \varepsilon$ with $\varepsilon \sim N(0, 1)$,

$$\Delta z = O(\sqrt{\Delta t}), \quad (\Delta z)^2 = O(\Delta t).$$

- Let us look at the quadratic term,

$$(\Delta X)^2 = a^2 (\Delta t)^2 + 2ab \Delta t \Delta z + b^2 (\Delta z)^2.$$

$\hookrightarrow$ the three terms have orders: $O((\Delta t)^2)$, $O((\Delta t)^{3/2})$, $O(\Delta t)$. As $\Delta t \downarrow 0$, the first two terms vanish.

- In the continuous-time limit, quadratic variation is then summarized by: $(dt)^2 = 0$, $dt dz_t = 0$, $\boxed{(dz_t)^2 = dt}$.
- If $dX_t = a dt + b dz_t$, then, to first order in dt ,

$$\boxed{(dX_t)^2 = b^2 dt.}$$

$\hookrightarrow$ The second-order Taylor term therefore survives:

$$\frac{1}{2} f_{XX} (dX_t)^2 = \frac{1}{2} b^2 f_{XX} dt.$$

Uncertainty affects the **drift** of a nonlinear transformation because Brownian variation accumulates at order dt .

Itô's Lemma

- Let the state variable x_t follow the diffusion model

$$dx_t = \mu_x(x_t) dt + \sigma_x(x_t) dz_t,$$

and consider a nonlinear transformation $y_t = f(x_t)$.

- From the second-order expansion,

$$dy_t = f_x dx_t + \frac{1}{2} f_{xx} (dx_t)^2.$$

and recall from the previous slide: $(dx_t)^2 = \sigma_x^2 dt$.

- Substituting both terms gives Itô's lemma:

$$dy_t = \left(\mu_x f_x + \frac{1}{2} \sigma_x^2 f_{xx} \right) dt + \sigma_x f_x dz_t.$$

→ A nonlinear transformation therefore changes both parts of the process:

$$\underbrace{\mu_y}_{\text{drift of } y} = \mu_x f_x + \frac{1}{2} \sigma_x^2 f_{xx}, \quad \underbrace{\sigma_y}_{\text{diffusion of } y} = \sigma_x f_x.$$

In particular, curvature f_{xx} affects the **expected change** in y .

- If the function also depends explicitly on time, $y_t = f(x_t, t)$, one additional term appears:

$$dy_t = \left(f_t + \mu_x f_x + \frac{1}{2} \sigma_x^2 f_{xx} \right) dt + \sigma_x f_x dz_t.$$

Itô Example: Simple Returns vs. Log Returns

- Suppose the risky asset follows $\frac{dS_t}{S_t} = \mu dt + \sigma dz_t$. Thus μ is the drift of the instantaneous simple return.
- We now ask how the log price evolves, $y_t = \log S_t$. For $f(S) = \log S$, we have: $f_S = \frac{1}{S}, f_{SS} = -\frac{1}{S^2}$.
- We start by the expansion: $d \log S_t = \frac{1}{S_t} dS_t - \frac{1}{2} \frac{1}{S_t^2} (dS_t)^2$. Since

$$dS_t = S_t(\mu dt + \sigma dz_t), \quad (dS_t)^2 = S_t^2 \sigma^2 dt,$$

substitution yields (Itô's lemma)

$$d \log S_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dz_t.$$

- The term $-\sigma^2/2$ comes entirely from the curvature of the logarithm: $f_{SS} < 0$.
 $\hookrightarrow$ Greater volatility therefore lowers expected log growth even though $\mathbb{E}_t \left[\frac{dS_t}{S_t} \right] = \mu dt$.
- $\hookrightarrow$ Simple and log returns therefore have different drifts:

$$\text{simple-return drift} = \mu, \quad \text{log-return drift} = \mu - \frac{1}{2} \sigma^2.$$

This is the continuous-time counterpart of the Jensen adjustment.

Dynamic Programming and the HJB

- The principle of dynamic programming is to break a dynamic optimization problem into a sequences. Let X_t denote the state and u_t the control. The value function summarizes the optimal attainable expected utility from the current state onward: ($\rho > 0$)

$$V(X_t, t) = \max_{\{u_s\}_{s \geq t}} \mathbb{E}_t \left[\int_t^\infty e^{-\rho(s-t)} U(X_s, u_s) ds \right].$$

- Over a short interval dt , Bellman's principle rewrites the same problem as

$$V(X_t, t) = \max_{u_t} \left\{ U(X_t, u_t) dt + e^{-\rho dt} \mathbb{E}_t [V(X_{t+dt}, t + dt)] \right\}.$$

→ HJB eq. is obtained by asking what this recursion implies as the interval dt becomes arbitrarily small.

- Once the current state is known, the agent faces the following trade-off in choosing u_t :

$$\underbrace{U(X_t, u_t)}_{\text{utility today}} \quad \text{and} \quad \underbrace{V(X_{t+dt}, t + dt)}_{\text{value of the future state}}.$$

From the Continuous-Time Problem to the HJB

- Let the state follow the diffusion model: $dX_t = \mu(X_t, u_t, t) dt + \sigma(X_t, u_t, t) dz_t$.
The control u_t can therefore affect both the expected evolution (μ) and the risk of the state (σ).
- From the previous slide:

$$V(X_t, t) = \max_{u_t} \left\{ U(X_t, u_t) dt + e^{-\rho dt} \mathbb{E}_t[V(X_{t+dt}, t+dt)] \right\}.$$

- For a short interval,

$$e^{-\rho dt} = 1 - \rho dt + o(dt).$$

- Once applied Itô's lemma to the continuation value $V(X_t, t)$:

$$dV = \left(V_t + \mu V_X + \frac{1}{2} \sigma^2 V_{XX} \right) dt + \sigma V_X dz_t.$$

- Since $\mathbb{E}_t[dz_t] = 0$, the expected change in continuation value is

$$\mathbb{E}_t[dV] = \left(V_t + \mu V_X + \frac{1}{2} \sigma^2 V_{XX} \right) dt.$$

→ We are now ready to substitute the discounting and continuation-value expansions into Bellman's.

From Bellman's Recursion to the HJB

- Hence

$$\mathbb{E}_t[V(X_{t+dt}, t+dt)] = V(X_t, t) + \left(V_t + \mu V_X + \frac{1}{2} \sigma^2 V_{XX} \right) dt + o(dt).$$

- Substitute these expressions into Bellman's recursion:

$$V = \max_{u_t} \left\{ U dt + (1 - \rho dt) \left[V + \left(V_t + \mu V_X + \frac{1}{2} \sigma^2 V_{XX} \right) dt \right] \right\} + o(dt).$$

- Keeping only terms of order dt ,

$$V = \max_{u_t} \left\{ V + \left[U + V_t + \mu V_X + \frac{1}{2} \sigma^2 V_{XX} - \rho V \right] dt \right\} + o(dt).$$

- Cancel V , divide by dt , and let $dt \downarrow 0$:

$$0 = \max_u \left\{ U(X, u) + V_t + \mu(X, u, t) V_X + \frac{1}{2} \sigma^2(X, u, t) V_{XX} - \rho V \right\}.$$

- If the problem is stationary:

$$\rho V = \max_u \left\{ U + V_t + \mu V_X + \frac{1}{2} \sigma^2 V_{XX} \right\}.$$

when V is not time dependent then $V_t = 0$.

→ The HJB is therefore Bellman's optimality condition written locally in continuous time.

How to Read the HJB

$$\rho V = \max_u \left\{ \underbrace{\quad}_{U} + \underbrace{\quad}_{V_t} + \underbrace{\quad}_{\mu V_X} + \underbrace{\quad}_{\frac{1}{2}\sigma^2 V_{XX}} \right\}$$

The equation shows the components of the HJB equation: current utility (U), time effect (V_t), expected state change (μV_X), and effect of local risk ($\frac{1}{2}\sigma^2 V_{XX}$).

How to read the HJB

- ρV is the required instantaneous return on the utility value represented by V .
- μV_X values the expected movement in the state at its marginal continuation value.
- $\frac{1}{2}\sigma^2 V_{XX}$ captures how uncertainty affects continuation value through curvature.
- For a concave value function (i.e., $V_{XX} < 0$), local uncertainty lowers continuation value.

What is missing?

- The HJB determines the shape of V in the **interior** of the state–time domain.
- To pin down a particular solution, we also need (economically appropriate) boundary/terminal conditions.
- In finite-horizon, the value function must match the payoff that remains at the terminal date ($V(X, T) = G(X)$).
 $\hookrightarrow$ HJB determines how this terminal value is propagated backward for all $t < T$.
- We may also need conditions at the boundaries of the state space. Depending on the economic problem, the boundary may pin down the level of V or impose a condition on its slope.

$\hookrightarrow$ The HJB gives the local optimality condition
VS. boundary/terminal, or later transversality conditions select the economically relevant solution.

The Hamiltonian: the Optimal Control

- The HJB can be written as: $\rho V = V_t + \max_{u \in \mathcal{U}} \{U(X, u) + \mu(X, u, t)V_X + \frac{1}{2}\sigma^2(X, u, t)V_{XX}\}.$
- The terms that depend on the control are collected in the **Hamiltonian**:

$$\mathcal{H}(X, t, u) = U(X, u) + \mu(X, u, t)V_X + \frac{1}{2}\sigma^2(X, u, t)V_{XX}.$$

- The optimal control therefore solves: $u^*(X, t) \in \arg \max_{u \in \mathcal{U}} \mathcal{H}(X, t, u).$
- If the optimum is an interior, the FOC is

$$\left. \frac{\partial \mathcal{H}}{\partial u} \right|_{u=u^*} = 0.$$

If the control is constrained, the maximization must respect the constraints as well.

- In the Merton problem, the controls will be consumption and the risky portfolio share. We will obtain their optimal policies directly from the corresponding Hamiltonian.

The Merton Consumption-Portfolio Problem

- Following Merton (1971), consider an investor who chooses how much to consume and how to allocate wealth between a risky asset and a risk-free asset over an infinite horizon.
↪ The investment opportunities are: $\frac{dA_t}{A_t} = rdt$, and $\frac{dS_t}{S_t} = \mu dt + \sigma dz_t$.
- At each date t , the investor chooses: i) the consumption rate, C_t , and ii) the fraction of wealth in the risky asset, ω_t . The remaining fraction $1 - \omega_t$ is invested in the risk-free asset.
- The objective is

$$V(W_t, t) = \max_{\{C_s, \omega_s\}_{s \geq t}} \mathbb{E}_t \left[\int_t^\infty e^{-\rho(s-t)} U(C_s) ds \right].$$

- The relevant state variable is financial wealth W_t .
The controls (C_t, ω_t) determine both the drift and the risk of its evolution.
- ↪ We therefore first derive the wealth dynamics implied by consumption and portfolio choice, and then insert them into the general HJB.

Wealth Dynamics

- With wealth W_t , the investor allocates

$\omega_t W_t$ to the risky asset, $(1 - \omega_t) W_t$ to the risk-free asset.

- Before consumption, the portfolio earns

$$\begin{aligned} & (1 - \omega_t) W_t r dt + \omega_t W_t (\mu dt + \sigma dz_t) \\ & = [r W_t + \omega_t W_t (\mu - r)] dt + \omega_t W_t \sigma dz_t. \end{aligned}$$

- Consumption is a flow variable: $C_t dt$ units of wealth over the interval dt .
- Hence the investor's wealth evolves according to

$$dW_t = [r W_t + \omega_t W_t (\mu - r) - C_t] dt + \omega_t W_t \sigma dz_t.$$

- Therefore the wealth drift and variance are

$$\mu_W = r W_t + \omega_t W_t (\mu - r) - C_t, \quad \sigma_W^2 = \omega_t^2 W_t^2 \sigma^2.$$

→ Consumption lowers expected wealth growth,
while risky investment affects both i) exp. wealth growth and ii) wealth risk.

The Merton HJB

- From the general continuous-time control problem,

$$\rho V(W, t) = \max_{C, \omega} \left\{ U(C) + V_t + V_W \mu_W + \frac{1}{2} V_{WW} \sigma_W^2 \right\}.$$

- From the wealth dynamics derived above,

$$\mu_W = rW + \omega W(\mu - r) - C, \quad \sigma_W^2 = \omega^2 W^2 \sigma^2.$$

- Substituting these expressions into the HJB gives

$$\rho V(W, t) = \max_{C \geq 0, \omega} \left\{ U(C) + V_t + V_W [rW + \omega W(\mu - r) - C] + \frac{1}{2} V_{WW} \omega^2 W^2 \sigma^2 \right\}.$$

- The two controls enter through different margins: C enters current utility and wealth drift, ω enters wealth drift and variance.
- Under the time-homogeneous infinite-horizon Merton benchmark, stationarity can be imposed later when solving for the value function.

CRRA Utility and Homogeneity

- Let us consider CRRA utility:

$$U(C) = \frac{C^{1-\gamma}}{1-\gamma}, \quad \gamma > 0, \quad \gamma \neq 1.$$

- Recall that γ determines both relative risk aversion ($-\frac{C U''(C)}{U'(C)}$) and the elasticity of intertemporal substitution.
- CRRA utility is homogeneous of degree $1 - \gamma$: $U(aC) = a^{1-\gamma}U(C)$, with $a > 0$.
- The wealth dynamics are linear in wealth and consumption. Hence, if a policy $\{C_s, \omega_s\}$ is feasible starting from wealth W , then $\{aC_s, \omega_s\}$ is feasible starting from wealth aW .
 $\hookrightarrow$ Scaling initial wealth by a scales the value of the entire optimization problem by $a^{1-\gamma}$:

$$V(aW, t) = a^{1-\gamma}V(W, t).$$

- This homogeneity property determines the dependence of the value function on wealth. We can therefore write

$$V(W, t) = \frac{1}{1-\gamma} W^{1-\gamma} v(t),$$

where the unknown function $v(t)$ collects the remaining dependence on time.

- The corresponding derivatives are

$$V_t = \frac{W^{1-\gamma}}{1-\gamma} v'(t), \quad V_W = W^{-\gamma} v(t), \quad V_{WW} = -\gamma W^{-\gamma-1} v(t).$$

Step-by-Step Solution: the CRRA Guess

- Start from the Merton HJB:

$$\rho V = \max_{C, \omega} \left\{ U(C) + V_t + V_W [rW + \omega W(\mu - r) - C] + \frac{1}{2} V_{WW} \omega^2 W^2 \sigma^2 \right\}.$$

- Under CRRA, substituting these expressions into the HJB gives

$$\rho \frac{W^{1-\gamma}}{1-\gamma} v(t) = \max_{C, \omega} \left\{ \frac{C^{1-\gamma}}{1-\gamma} + \frac{W^{1-\gamma}}{1-\gamma} v'(t) + W^{-\gamma} v(t) [rW + \omega W(\mu - r) - C] - \frac{1}{2} \gamma W^{1-\gamma} v(t) \omega^2 \sigma^2 \right\}.$$

- Divide both sides by $W^{1-\gamma}/(1-\gamma)$:

$$\rho v(t) = \max_{C, \omega} \left\{ \left(\frac{C}{W} \right)^{1-\gamma} + v'(t) + (1-\gamma) v(t) \left[r + \omega(\mu - r) - \frac{C}{W} \right] - \frac{1}{2} \gamma (1-\gamma) v(t) \omega^2 \sigma^2 \right\}.$$

→ Homogeneity removes the level of wealth: the choice of consumption now enters only through the ratio C/W .

Step-by-Step HJB Solution: Optimal Controls

- From the normalized HJB, the terms involving consumption are

$$\left(\frac{C}{W}\right)^{1-\gamma} - (1-\gamma)v(t)\frac{C}{W}.$$

The first-order condition for C is therefore

$$0 = \frac{\partial}{\partial C} \left[\left(\frac{C}{W}\right)^{1-\gamma} - (1-\gamma)v(t)\frac{C}{W} \right] = \frac{1-\gamma}{W} \left[\left(\frac{C}{W}\right)^{-\gamma} - v(t) \right] \iff \frac{C^*}{W} = v(t)^{-1/\gamma}.$$

↪ Optimal consumption is therefore proportional to wealth, with a consumption–wealth ratio governed by $v(t)$.

- The terms involving the portfolio share are

$$(1-\gamma)v(t) \left[\omega(\mu-r) - \frac{1}{2}\gamma\omega^2\sigma^2 \right].$$

The first-order condition for ω gives

$$(\mu-r) - \gamma\omega\sigma^2 = 0 \iff \omega^* = \frac{\mu-r}{\gamma\sigma^2}$$

↪ The risky share is already determined by investment opportunities and risk aversion, while the consumption rule still depends on the unknown function $v(t)$.

▷ We now substitute both optimal controls back into the HJB to determine the evolution of $v(t)$.

Step-by-Step HJB Solution: Back to the HJB

- At the optimal consumption choice,

$$\left(\frac{C^*}{W}\right)^{1-\gamma} - (1-\gamma)v(t)\frac{C^*}{W} = v(t)^{1-1/\gamma} - (1-\gamma)v(t)^{1-1/\gamma} = \gamma v(t)^{1-1/\gamma}.$$

- At the optimal portfolio choice,

$$\omega^*(\mu - r) - \frac{1}{2}\gamma(\omega^*)^2\sigma^2 = \frac{(\mu - r)^2}{2\gamma\sigma^2}.$$

- Substituting both optimal controls into the normalized HJB gives

$$\rho v(t) = v'(t) + (1-\gamma) \left[r + \frac{(\mu - r)^2}{2\gamma\sigma^2} \right] v(t) + \gamma v(t)^{1-1/\gamma}.$$

- Equivalently,

$$v'(t) = \left[\rho - (1-\gamma)r - (1-\gamma)\frac{(\mu - r)^2}{2\gamma\sigma^2} \right] v(t) - \gamma v(t)^{1-1/\gamma}.$$

- The optimization over (C, ω) is done. What remains is a (nonlinear) differential equation for the time component $v(t)$ of the value function.

Step-by-Step HJB Solution: Linearizing the ODE

- From the previous slide,

$$v'(t) = a v(t) - \gamma v(t)^{1-1/\gamma},$$

where $a := \rho - (1 - \gamma)r - (1 - \gamma)(\mu - r)^2 / (2\gamma\sigma^2)$.

- The equation is nonlinear because of the term $v(t)^{1-1/\gamma}$. To simplify it, define

$$y(t) := v(t)^{1/\gamma} \quad \Longleftrightarrow \quad v(t) = y(t)^\gamma.$$

- Differentiating,

$$v'(t) = \gamma y(t)^{\gamma-1} y'(t).$$

- Substituting $v(t) = y(t)^\gamma$ and $v'(t) = \gamma y(t)^{\gamma-1} y'(t)$ into the ODE gives

$$\gamma y^{\gamma-1} y' = a y^\gamma - \gamma y^{\gamma-1}.$$

- Dividing by $\gamma y^{\gamma-1}$ yields the linear equation

$$y'(t) = \frac{a}{\gamma} y(t) - 1.$$

→ The transformation $y(t) = v(t)^{1/\gamma}$ converts the nonlinear HJB equation into a first-order linear ODE.

- ▷ In the stationary Merton benchmark, the time component is constant, so the next step is to impose $y'(t) = 0$ and recover a constant consumption–wealth ratio.

Closed-Form and Stationary Solution

- For $a \neq 0$, the linear ODE, $y'(t) = \frac{a}{\gamma}y(t) - 1$, has the general solution: $y(t) = \frac{\gamma}{a} + (y(0) - \frac{\gamma}{a})e^{\frac{a}{\gamma}t}$.
Since $y(t) = v(t)^{1/\gamma}$,

$$v(t) = \left[\frac{\gamma}{a} + \left(v(0)^{1/\gamma} - \frac{\gamma}{a} \right) e^{\frac{a}{\gamma}t} \right]^\gamma.$$

- In the stationary infinite-horizon Merton problem, the value function does not depend on calendar time. Hence we select the constant solution:

$$y(t) \equiv \bar{y} \quad \Longleftrightarrow \quad y'(t) = 0.$$

- From the ODE, $0 = \frac{a}{\gamma}\bar{y} - 1 \implies \bar{y} = \frac{\gamma}{a}$,
which is equivalently obtained from the general solution by setting: $y(0) = \gamma/a$.
- Therefore

$$\bar{v} = \left(\frac{\gamma}{a} \right)^\gamma, \quad \frac{C_t^*}{W_t} = \bar{v}^{-1/\gamma} = \frac{a}{\gamma} \equiv \kappa.$$

- The general ODE allows a time-varying value-function coefficient;
stationarity selects its constant solution and therefore a constant consumption–wealth ratio.

The Stationary Merton Solution

- Recall: $a := \rho - (1 - \gamma)r - (1 - \gamma) \frac{(\mu - r)^2}{2\gamma\sigma^2}$.
- The stationary value-function coefficient is: $\bar{v} = \left(\frac{\gamma}{a}\right)^\gamma$.
- The two optimal policies are therefore

$$\frac{C_t^*}{W_t} = \kappa = \frac{a}{\gamma},$$

$$\omega_t^* = \frac{\mu - r}{\gamma\sigma^2}.$$

- Equivalently, the optimal consumption–wealth ratio is

$$\kappa = \frac{1}{\gamma} \left[\rho - (1 - \gamma)r - (1 - \gamma) \frac{(\mu - r)^2}{2\gamma\sigma^2} \right].$$

- Hence both optimal policies are **scale free**:

$$C_t^* = \kappa W_t, \quad \omega_t^* = \text{constant}.$$

Wealth changes the level of consumption and risky investment, but not their proportions relative to wealth.

- A stationary solution with positive consumption requires

$$\kappa > 0 \iff a > 0,$$

or equivalently

$$\rho > (1 - \gamma)r + (1 - \gamma) \frac{(\mu - r)^2}{2\gamma\sigma^2}.$$

→ The Merton problem therefore delivers two simple rules:

- i) a constant fraction of wealth is consumed, and ii) a constant fraction is invested in the risky asset.

TBD

Bibliography I

John Y. Campbell. *Financial Decisions and Markets: A Course in Asset Pricing*. Princeton University Press, Princeton, NJ, 2018.

John H. Cochrane. *Asset Pricing*. Princeton University Press, Princeton, NJ, revised edition, 2005.

Robert C. Merton. Optimum consumption and portfolio rules in a continuous-time model. *Journal of Economic Theory*, 3(4):373–413, 1971. doi: 10.1016/0022-0531(71)90038-X.